

NSE / BSE SCANNER SUITE

COMPLETE SPECIFICATION — BOTH SYSTEMS

Final Locked Version • June 2026

SYSTEM 1	VS	SYSTEM 2
Momentum Scanner v2.0		Setup Discovery Engine
Assumption-Based		Data-Driven
Conviction Levels		Backtest Rankings
Live Signals Now		Evidence First
Your best guess at what works		Let data tell you what actually works

Both systems share: Stock universe • Hard filters • Global bleeding filter • Research journal
Telegram delivery • News scanner • Dedup registry • Weekly + quarterly reports
Phase 1: GitHub Actions • yfinance • pandas-ta • Telegram Channel • Phase 2: Zerodha Kite API

SYSTEM 1

Conviction-Based Opportunity Discovery Engine

Your current scanner — assumption-based setups — generates live signals now

Purpose: find opportunities, provide full context, YOU decide to trade

SYSTEM 1 — TABLE OF CONTENTS

Philosophy	Why we simplified — conviction over fake precision
Layer 1	Data Sources — Weekly + Daily + Hourly + Global
Layer 2	Hard Filters — 5 sequential layers
Layer 3	Weekly Structural Gate + Sector Rotation
Layer 4	Daily Signal Generation — 3 questions
Layer 5	Four Conviction Levels — replaces scoring
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Arch 1	GitHub Architecture — System 1

SYSTEM 1 — PHILOSOPHY

Core Insight — Why We Simplified

OLD PROBLEM: RSI rising = +7pts | MACD cross = +5pts | Why 7? Why 5? No answer. Points were invented not data-driven. 84% probability looked precise but was made up. NEW APPROACH: Binary/ternary decisions at each layer 4 conviction levels replace 110-point fake scoring 3 honest probability bands (not layered arithmetic) Calibrated from ACTUAL outcomes after 3 months tracking SYSTEM PURPOSE: Find technically valid setups on 450+ NSE stocks Provide full market context + execution guidance YOU make the final decision to trade or not Research journal learns from every outcome

What Changed vs What Stayed

Removed	Replaced With
110-point fake scoring	4 conviction levels (High/Moderate/Low/Watchlist)
Arbitrary probability layers	3 honest bands (70-80% / 55-65% / 40-50%)
Sharpe/Sortino in scoring	Weekly report only — informational
Complex news sentiment scoring	Binary flag + advisory for final 15 only
6-month backtest	3-year backtest covering all market phases
Fake Rs.1Cr capital + P&L	Pure signal outcome tracking — research journal
Position sizing in rupees	Removed — you decide your own sizing

LAYER 1 — DATA SOURCES

Timeframe	Purpose	Period	Interval	Source
Weekly	Structural trend — eligibility gate	1 year	1wk	yfinance batch
Daily	Signal generation — primary	6 months	1d	yfinance batch
Hourly	Execution planning — entry/stop/R:R	60 days	1h	yfinance batch
Index/Sector	Market regime + sector status	3 months	1d	yfinance
OI Data	FNO analysis	Daily	EOD	NSE bhav copy (after 6 PM)
Global	US futures + Asian markets bleeding	Daily	Live	yfinance
GIFT Nifty	India pre-market absolute override	Daily	Live	investing.com
News (market)	World top 3 + India top 10 headlines	Daily	Cached	Google RSS
News (stocks)	Final 15 selected stocks only	Daily	30 days	Google RSS + NSE

Universe — 450+ NSE Stocks

Index	Cap	Count	Priority
Nifty 50	Large	50	1st
Nifty 150	Large	150	1st
Nifty Midcap 150	Mid	150	2nd
Nifty Smallcap 250	Small	250	3rd
Total deduped	Mixed	~450	Large > Mid > Small

Batch + Hourly Strategy

450 stocks → 9 batches of 50 → ONE yfinance call per batch Weekly: 9 | Daily: 9 | Hourly: 9 = 27 batch calls + ~5 index calls = ~32 total Hourly strategy: >=4 candles today = use today | <4 candles = last 5 trading days 8 AM run: previous full day hourly data used

LAYER 2 — HARD FILTERS (5 SEQUENTIAL LAYERS)

All filters before indicator calculation. Hard blocks cannot be overridden by conviction. Early rejection saves compute.

L1	Weekly Gate Close < 20W EMA → EXCLUDE entirely ~30-40% removed. No further processing.
L2	Corporate Actions ±1 trading day of event → BLOCKED Results/Dividend/Split/Bonus/Buyback/Rights. NOT: AGM, EGM.
L2B	Sector Bleeding ★ Sector index < -1.5% on day → ENTIRE SECTOR BLOCKED Independent of Nifty. Exception: FNO Long Unwinding still generated.
L3	Global Bleeding GIFT Nifty < -1.5% OR 70%+ of 8 indices < -1.5% → BLEEDING GIFT = absolute override. Only FNO Long Unwinding allowed in bleeding.
L4	Liquidity + Quality Avg vol < 50k OR price < Rs.20 OR < 50 days data → SKIP Prevents illiquid/penny/incomplete data signals.

Nifty Regime — Signal Permissions

Regime	Momentum	Reversal A+B	FNO C1	FNO C2
Strong Bull / Bull	YES	YES	YES	Warning only
Weak Bull / Neutral	YES	YES	YES	Warning only
Weak Bear	NO ■	YES	YES	ACTIVE signal ■
Strong Bear	NO ■	NO ■	YES	ACTIVE signal ■

Global Bleeding Basket

US Futures (3): ES=F NQ=F YM=F Asian (5): ^N225 ^HSI ^KS11 ^AXJ0 000001.SS 70% threshold: 6 of 8 must breach -1.5% for basket rule GIFT Nifty checked FIRST — absolute override regardless of basket

- Sector bleeding: independent of Nifty — surgical sector block
- Sector bleeding exception: FNO Long Unwinding generated even in bleeding sector
- GIFT Nifty < -1.5% = immediate bleeding — checked before basket calculation
- Sector bleeding applied retroactively in 3-year backtest

LAYER 3 — WEEKLY STRUCTURAL GATE + SECTOR ROTATION

Status	Condition	Conviction Cap	Action
ELIGIBLE	Close clearly > 20W EMA (>2% above)	HIGH possible	Proceed to daily scan
MARGINAL	Close > 20W EMA but within 2%	MODERATE maximum	Proceed — capped
EXCLUDED	Close < 20W EMA	None	Skip entirely

Sector Rotation Detection

Definition: sector was BELOW 20D EMA last week, now ABOVE 20D EMA this week Action: stocks in rotating sector get ranking priority eligible for ONE conviction level upgrade (max HIGH) Tag: ■ SECTOR ROTATING UP — [SectorName] Header: ■ ROTATING UP THIS WEEK: IT sector, PSU Banking

LAYER 4 — DAILY SIGNAL GENERATION

Three Core Questions — Binary/Ternary Output Only

Question	YES Condition	Signal Type
Q1: Breakout?	Close > 20D High OR W-Pattern neckline OR BB upper AND Vol >= 1.3x AND Supertrend bullish (10,3)	Momentum
Q2: Reclaim valid?	9 EMA crossed ABOVE 16 EMA FRESH today (not already crossed) AND Close > 20 EMA AND Vol >= 1.3x OR: Retest recovery (prior breakout + shakeout + recovery above + fresh cross)	Reversal
Q3: Momentum?	EXPANDING: RSI rising 2+ bars + MACD hist growing + Vol above avg NEUTRAL: RSI rising 1 bar OR vol at average CONTRACTING: RSI falling OR MACD shrinking OR vol below avg	Quality filter

Momentum Patterns — Phase 1 (3 Only)

1. 20D HIGH: Close > highest high of last 20 bars 2. W-PATTERN: 30 bar lookback daily+weekly | tolerance: Large 2.5% Mid 3.5% Small 5.0% Neckline breakout | Daily+Weekly both = max conviction 3. BB BREAKOUT: Close > upper BB (20,2) All require: Close > 20 EMA + Volume >= 1.3x + Supertrend bullish (10,3) FLAG PATTERN: Phase 2 after backtest data collected

Reversal Setups

Setup A — EMA Cross: fresh 9x16 cross today + Close>20EMA + Vol>=1.3x Setup B — Retest Recovery: prior breakout + shakeout below + recovery above + fresh 9x16 cross + Close>20EMA + Vol>=1.3x Setup C1 — Short Covering (FNO): OI falling>5% + price rising + PCR>1.0 + Close>20EMA Setup C2 — Long Unwinding (FNO): OI falling + price falling Bull: blocks longs | Weak/Strong Bear: ACTIVE signal

Signal Timeframe Validity (shown on every card)

Pattern	T1	T2	Max Hold
20D High	3-10d	5-15d	15 days
W Daily	5-15d	10-20d	20 days
W Weekly/Both	10-20d	20-30d	30 days
BB Breakout	2-7d	5-12d	12 days
EMA Cross A	3-10d	5-15d	15 days
Retest B	5-15d	10-20d	20 days
Short Covering C1	1-5d	3-8d	8 days
Long Unwinding C2	1-3d	—	3 days

LAYER 5 — FOUR CONVICTION LEVELS

■ Replaces 110-point scoring. Conviction = weekly structure + daily questions + market context. Binary logic not arithmetic.

HIGH ■■■■

Prob: 70-80%

Weekly: strong structure above 20W EMA | Daily: ALL 3 questions YES Market: Bull or Strong Bull | Sector: above 20D EMA or rotating up Momentum Q3: EXPANDING for 2+ bars

MODERATE ■■■

Prob: 55-65%

Weekly: eligible (marginal ok) | Daily: 2 of 3 questions YES Market: Neutral or better | Sector: not bleeding Momentum Q3: NEUTRAL or EXPANDING

LOW ■■

Prob: 40-50%

Weekly: eligible | Daily: 1 of 3 questions YES Market: not Strong Bear | Sector: not bleeding Momentum: any | Shown in separate Watchlist section

WATCHLIST ■

Prob: < 40%

Signal exists but conditions marginal DO NOT TRADE — monitor for upgrade next scan Separate section in Telegram footer

Conviction Upgrade Rules (max 1 per signal, no stacking)

Trigger	Effect	Maximum
Sector rotating up this week	+1 level	HIGH
W-Pattern daily + weekly both confirmed	+1 level	HIGH
FNO Long Buildup (OI rising + price rising)	+1 level	HIGH
Marginal weekly (within 2% of 20W EMA)	CAP at MODERATE	MODERATE
Strong negative news	-1 level downgrade	—

Probability Bands — Honest Estimates

These are starting estimates – calibrated from actual outcomes after 3 months tracking HIGH: 70-80% MODERATE: 55-65% LOW: 40-50% Quarterly report recalibrates based on real win rates e.g. after 3 months: "HIGH W-Pattern: 73% actual" replaces estimate

LAYER 6 — HOURLY EXECUTION PLAN

Generated for HIGH and MODERATE only. Phase 1 = manual decision aid on phone. Phase 2 = Zerodha auto-execution input.

Output	How Calculated	Action Signal
Entry Zone	Hourly 8 EMA + 20 EMA Low of last 2 candles if consolidating near EMA	Near EMA = good entry
Stop Placement	ATR(14)×1.5 below entry OR nearest hourly swing low — use tighter	Both shown — you choose
Nearest Resistance T1	Prior hourly high (last 5 sessions) OR daily T1	Intraday target
Risk/Reward	(T1-entry)/(entry-stop) >=2.0 EXCELLENT 1.5-2.0 GOOD <1.5 WAIT	Min 1.5 to act
Timing Windows	GOOD: 10AM-11:30AM + 1PM-2:30PM AVOID: 9:15-10AM + 2:30-3:30PM	Current time shown

Hourly Data Strategy

Scan Time	Candles Today	Data Used
8 AM	0 (pre-market)	Previous full trading day
10 AM / 11:30 AM	2-3 candles	Last 5 trading days (25-30 candles)
1 PM onwards	4+ candles	Today's data (threshold met)

LAYER 7 — SECTOR DISTRIBUTION + LAYER 8 — RISK & EXIT

Sector Distribution Cap

Maximum 2 stocks from same sector per signal type per run Prevents: all 5 signals from same sector when that sector reverses 3rd+ from same sector → Watchlist section (labelled sector overflow) Rotation priority: rotating sectors ranked first within same conviction level

Exit Structure — All Types

Exit	Trigger	Action	Research Label
SL Hit	Price hits stop loss	Full exit immediately	SL_HIT
T1 Hit	Price reaches T1	Exit 50% + trail SL to entry	T1_HIT
T2 Hit	Price reaches T2	Exit remaining 50%	T2_HIT
Trailing	After T1: trail to prev day low	Update daily	TRAIL_EXIT
Expired	Max hold days reached	Exit at market	EXPIRED
Structure Break	Close below 20 EMA daily (MOMENTUM ONLY)	Exit	STRUCTURE_EXIT
Tier 2 Sentiment	Sector bleeding OR regime -2 levels	Exit recommendation	SENTIMENT_EXIT
Tier 3 Urgent	Price below 20EMA + RSI fall + high vol	Urgent exit alert	TREND_EXIT

■ Structure break exit: momentum signals only. Reversals exempt — designed to catch recoveries.

LAYER 9 — RESEARCH JOURNAL + FAILURE ANALYSIS

■ No fake capital. No imaginary P&L.; Pure signal outcome research — did the setup work? Under what conditions? Why did it fail? How to improve?

Every Signal Recorded

data/research_journal/signal_master.csv Key columns: date, scan_time, symbol, sector, cap_type, signal_type, pattern, conviction_level, probability_band, entry_suggested, sl, t1, t2, rr_ratio, market_regime, sector_status, sector_rotating, news_flag, global_status, q1, q2, q3, oi_pattern Outcomes (updated Sunday): price_t1_hit, price_t2_hit, price_sl_hit, max_gain_pct, max_loss_pct, price_5d, price_10d, price_20d, outcome_label, failure_category, days_to_outcome

Pattern Failure Analysis — 5 Categories

Category	Common Factors	Typical Fix Suggested
False Breakout	Volume faded day 2 Near 52W high Neutral regime	Require 2-day close above level 52W proximity check
EMA Cross Fail	Vol at minimum RSI < 45 on cross Neutral market	Vol >= 1.5x in neutral RSI > 45 mandatory
W-Pattern Fail	Lows unequal Neckline at resistance Weak weekly	Tighten tolerance Neckline vs resistance check
Timing Failure	In avoid window Friday late Pre-event day	Strengthen window rules Event calendar check
Sector Drag	Sector near 20D EMA Deteriorated post-signal	Tighten sector threshold Require clearly above EMA

Weekly Report — 8 Messages (Sunday 10 AM)

Msg	Content
1	Opportunity summary: signals by conviction/type/sector this week
2	Outcome tracking: prior signals resolved — T1/T2/SL/expired rates
3	Pattern performance: win rate by pattern, conviction, regime
4	Failure analysis: categories, common factors, improvement suggestions
5	Conviction calibration: are HIGH signals actually outperforming?
6	Sector analysis: best/worst sectors, rotation outcomes
7	3-year backtest snapshot
8	Next week watchlist: stocks near key levels, building setups

Quarterly — Every 13th Sunday

System performance | Pattern deep dive | Failure pattern library Proposed upgrades with evidence + impact estimate | Alpha vs Nifty benchmark Conviction band recalibration from actual win rates

PARALLEL PROCESSING + NEWS SCANNER

Parallel Processing — Target < 4 Min/Run

Level	Method	For	Workers
1	yfinance batch	50 stocks per call — 27 batches (W+D+H)	27 calls
2a	ThreadPoolExecutor	News + OI + GIFT + global (I/O bound)	20 threads
2b	ProcessPoolExecutor	Indicators + signals + conviction + execution (CPU)	4 processes
3	Pipeline	Pre-fetch→filter→process→rank→news→deliver	Staged

Stage 1 Pre-fetch (~100s): W+D+H batches parallel + sectors + global + corp actions Stage 2 Filter (~5s): 5 layers → ~150-200 pass | sector cap applied | dedup applied Stage 3 Process (~60s): ThreadPool news + ProcessPool indicators run simultaneously Stage 4 Execute (~15s): Execution plans HIGH+MOD only Stage 5 Rank (~5s): Top 5 per type | sector overflow to watchlist Stage 6 News (~15s): Fetch news for FINAL 15 stocks only Stage 7 Deliver (~10s): 5 Telegram messages + update registries TOTAL: ~210 seconds (~3.5 min) ■ | Monthly: ~26×3.5=~390 min (~20% of 2000 free tier)

News Scanner — Two Parts

Part	What	For Which Stocks	Purpose
A — Stock News	Last 30 days: corporate events + sentiment	Final 15 ranked stocks ONLY	Conviction builder for YOUR decision
B — Market Headlines	World top 3 + India top 10	All scans (header message)	Broad market context

PART A – Advisory not auto-filter: Signal selected on PURE TECHNICALS first News checked AFTER selection for final 15 ■ Strengthens conviction | ■ Review before trading | ■ Pure technical NOT auto-downgrading – YOUR judgment wins PART B – Market headlines cached at 8AM, refreshed at 1PM + 3PM 80 char max per headline | No URLs | Fails silently

SCHEDULE + TELEGRAM DELIVERY

Run	Time IST	Days	Purpose
1	8:00 AM	Mon-Fri	Pre-market: global + GIFT + rotation + watchlist (no signals)
2	10:00 AM	Mon-Fri	Market open scan — all signals
3	11:30 AM	Mon-Fri	Mid morning scan — all signals
4	1:00 PM	Mon-Fri	Midday scan — all signals
5	3:00 PM	Mon-Fri	Pre-close scan — all signals
6	10:00 AM	Sunday	Weekly research report — 8 messages

Cron (UTC = IST - 5:30): - cron: '30 2 * * 1-5' # 8AM | - cron: '30 4 * * 1-5' # 10AM - cron: '0 6 * * 1-5' # 11:30 | - cron: '30 7 * * 1-5' # 1PM - cron: '30 9 * * 1-5' # 3PM | - cron: '30 4 * * 0' # Sunday 10AM workflow_dispatch: true # manual trigger always available

5 Messages Per Scan Run

Msg	Content	Signals
1	Header: regime + sector pulse + rotating sectors + market news (W3+I10)	—
2	Top 5 Momentum Breakout — by conviction then R:R — any cap	5
3	Top 5 Reversal (A+B) — by conviction then R:R — any cap	5
4	Top 5 FNO (C) — by conviction — FNO stocks only	5
5	Footer: emoji guide + LOW/WATCHLIST section + global mini summary	watchlist

Signal Card Format

[■/■/■] SYMBOL [■/■/■] Cap | Sector [■ FULL ALIGNMENT] or [■ SECTOR ROTATING UP] ■■■■ HIGH CONVICTION (or ■■■ / ■■)
■ Signal Type – Pattern | ■ Conditions triggered ■ Rs.Price ■ Vol:Xx ■ OI:DD-Mon(FNO) ■ Buy: Rs.XX ■ SL: Rs.XX ■ T1: Rs.XX ■Xd ■ T2: Rs.XX ■Xd ■ Prob: 70-80% (HIGH) ■ EXECUTION: Entry Rs.XX-XX | Stop Rs.XX | T1 Rs.XX | R:R X.X ■ | Timing: GOOD ■ NEWS (last 30d): ■ Order win Rs.420Cr (8d ago) OR ■ SEBI notice (12d ago) OR ■ Pure technical ■ Commentary

SYSTEM 1 — GITHUB ARCHITECTURE

```
momentum-scanner/ (PUBLIC repo) ■■■■ .github/workflows/scanner.yml ■■■■ data/ ■■■■ universe/ nifty50.csv nifty150.csv
midcap150.csv smallcap250.csv fno_stocks.csv ■■■■ sectors/ sector_map.csv ■■■■ corporate_actions/ actions.csv
(update weekly Sunday) ■■■■ holidays/ nse_holidays_2025.csv nse_holidays_2026.csv ■■■■ active_signals/
active_signals.csv ■■■■ backtest/ backtest_results.csv backtest_summary.csv ■■■■ research_journal/
signal_master.csv signals_YYYY_WW.csv outcomes.csv failure_analysis.csv ■■■■ src/ ■■■■ universe.py data_fetcher.py
filters.py sector_bleeding.py global_markets.py ■■■■ indicators.py (W+D+H) weekly_gate.py market_regime.py ■■■■
momentum_breakout.py reversal_breakout.py fno_signals.py ■■■■ conviction.py execution_plan.py sector_distribution.py
■■■■ risk_manager.py sentiment_monitor.py signal_registry.py ■■■■ news_scanner.py parallel_runner.py ■■■■
backtester.py research_journal.py ■■■■ failure_analyser.py conviction_calibrator.py ■■■■ weekly_reporter.py
quarterly_reporter.py ■■■■ telegram_bot.py ■■■■ main.py weekly_report.py requirements.txt README.md
```

GitHub Secrets

```
TELEGRAM_BOT_TOKEN + TELEGRAM_CHAT_ID (channel ID starting -100...)
```

Maintenance

Frequency	Task	Time
Quarterly	Update universe CSVs + FNO list	15 min
Weekly Sun	Add upcoming corp actions	5 min
Annually	Update trading holidays	5 min

System 1 Rating

Component	Rating	Notes
Signal logic + 3 questions	7.5/10	Solid foundation
Three-timeframe W+D+H	8.5/10	Hourly adds professional edge
Conviction system	8.0/10	Honest vs fake points
Parallel processing	8.0/10	<4 min for 450 stocks
Research journal + failure analysis	8.5/10	Data-driven improvement path
OVERALL	7.8/10	High Advanced Retail approaching Professional

SYSTEM 2

Data-Driven Setup Selection via Backtesting

Let the DATA tell you what works — backtest first, build second

Evidence selects the setups — not gut feeling or assumption

SYSTEM 2 — TABLE OF CONTENTS

Philosophy	Why data-driven — the problem with assumption-based setups
What Gets Tested	All EMA crosses + breakout patterns + candle patterns
Backtest Method	2 years × 500 stocks × no lookahead bias
Ranking Output	Master ranking table + detailed report per setup
Slot Architecture	4 preserved slots — 2 live + 2 research
Live Scanner	Signals from top 2 data-selected setups
Research Zone	Slots 3+4 tracked silently — ready to promote
Reports	Weekly + monthly + quarterly + re-rank
Arch 2	GitHub Architecture — System 2

SYSTEM 2 — PHILOSOPHY

The Core Problem System 2 Solves

System 1 problem: "I think 9x16 EMA cross works" – ASSUMPTION not FACT "I think W-pattern is reliable" – GUT FEELING not DATA System 2 solution: Let the data tell you what works Backtest every setup combination on 500 stocks × 2 years Evidence selects the setups – not gut or suggestion This is how quant funds start: TEST FIRST. BUILD SECOND. After backtest: Top 2 ranked setups → go into live scanning (System 2 scanner) Next 2 ranked setups → go into research zone (tracked silently) You compare: does live performance match backtest prediction? Quarterly: re-rank, tweak top 2, promote research setups if ready

Three Phases

Phase	Name	When	Output
1	Discovery	Now — build + run backtest	Full ranking of 47+ setups. You select top 4.
2	Selection	After first report	Top 2 → live signals. Next 2 → research zone.
3	Live + Research	Ongoing weekly	Top 2 signal cards to Telegram. Research tracked silently. Monthly/quarterly review.

WHAT GETS BACKTESTED

Category 1 — EMA Crosses (11 combinations)

EMA Pair	Timeframe	With Volume	Without Volume
5×10, 5×20, 8×21	Daily	YES (1.3x, 1.5x, 2x)	YES
9×16, 9×20, 9×21	Daily + Weekly	YES	YES
10×20, 10×50	Daily	YES	YES
20×50, 20×200, 50×200	Daily + Weekly	YES	YES

Category 2 — Breakout Patterns (5 types)

Pattern	Variations Tested
N-Day High Breakout	10D / 20D / 30D / 50D / 52-Week high
Bollinger Band	(20,2) and (20,2.5) — with/without trend filter
Volume Breakout	Price + vol > 1.3x / 1.5x / 2x average
Range Breakout	Tight consolidation (ATR contraction) + expansion
Weekly Breakout	Close > prior week high — with/without volume

Category 3 — Candle Patterns (8 types)

Pattern	Condition Variations
Bullish Engulfing	At 20 EMA / at prior low / with volume
Hammer / Inverted Hammer	At support / with volume confirmation
Morning Star	Daily / weekly timeframe
Piercing Line	At support zone
3 White Soldiers	After downtrend / after consolidation
Doji Reversal	At key levels / with volume
Inside Bar Breakout	Daily / with trend filter
Pin Bar	At EMA / at S/R level

Category 4 — Combined Conditions

After individual testing – best combinations tested: Best EMA cross + volume condition Best breakout + trend filter (above 20W EMA) Best candle + EMA position (price above/below EMAs) These become the final setup candidates for ranking

BACKTEST METHODOLOGY

Key Parameters

Parameter	Value	Reason
Universe	~500 NSE stocks	Nifty 50+150 + Midcap150 + Smallcap250
Period	2 years rolling	Covers multiple market phases
Data	Daily OHLCV + Weekly	yfinance period='2y' — free
Lookahead bias	NONE	Signal on day D using D-1 data only
SL method	ATR(14) × 1.5 below entry	Consistent across all setups
T1	1.5R (1.5 × risk)	Consistent across all setups
T2	2.5R (2.5 × risk)	Consistent across all setups
Regime tag	Bull/Neutral/Bear per signal	Enables regime-filtered analysis
Sector tag	Sector above/below 20D EMA	Enables sector analysis
First run	Manual — 20-40 min	One-time comprehensive test
Quarterly update	Rolling 2yr rerun	New data replaces oldest

What Gets Measured Per Setup

For each setup × each stock × each signal generated: win_rate_1_5R: % of signals where price reached T1 (1.5R)
 win_rate_2_5R: % of signals where price reached T2 (2.5R) avg_days_to_t1: how quickly T1 was hit (speed matters)
 avg_max_drawdown: how much it dropped before recovering false_signal_rate: % where price reversed immediately
 signal_frequency: how many signals per year (too rare = unusable) regime_performance: win rate in Bull / Neutral / Bear
 separately cap_performance: win rate for Large / Mid / Small separately REGIME INDEPENDENCE SCORE: Most valuable setups
 work across ALL regimes Setup that works 72% Bull but 41% Neutral = regime-dependent Setup that works 65%+ in ALL
 regimes = regime-independent Regime-independent setups rank HIGHER in final ranking

Market Regime Tagging Per Signal

Every signal tagged with market conditions at time of generation: Nifty 50 vs 50 DMA → Bull / Neutral / Weak Bear /
 Strong Bear Sector vs 20D EMA → Above / Near / Below Volume context → High (>1.5x) / Normal / Low This allows: "Setup
 X works 79% when market Bull + sector above 20D EMA" "Setup X works only 38% in Neutral regime — avoid then"

RANKING OUTPUT

Master Ranking Table (Telegram + CSV)

SETUP DISCOVERY ENGINE — BACKTEST RESULTS 500 stocks × 2 years | Setups tested: 47 | Total signals: ~125,000 RANK SETUP

WIN_1.5R	WIN_2.5R	AVG_DAYS	REGIME_INDEP	FREQ/YR	
					1 20D
High + Vol 1.5x	68%	41%	6.2d YES (62%+)	1,420	2 9×21 EMA Cross + Vol 1.5x 64% 38% 7.8d YES (60%+)
890	3	50D	High + Trend filter 63%	35%	9.1d NO (bull) 620
4	Engulfing at 20	EMA 61%	33%	5.4d NO (bull) 380	5 Bollinger breakout (20,2)
59%	31%	6.8d YES (57%+)	760	6 20×50 EMA Cross 58%	29%
11.2d YES (55%+)	440	7 Hammer at 20	EMA + Vol 57%	28%	5.1d YES (54%+)
520	8	W-Pattern neckline 71%	48%	12.4d NO (rare) 180	... 47 5×10 EMA Cross 38%
18%	3.1d NO	2,200			
RECOMMENDATION: Select setups from top of list Prefer regime-independent setups for live scanning High frequency + regime-independent = most reliable scanner					

Why W-Pattern Ranks #8 Despite 71% Win Rate

W-Pattern: 71% win rate BUT: Very few signals per year (180 vs 1,420 for 20D High) Long holding period (12+ days average) Only works in bull regime (regime-dependent) Hard to detect reliably in automated code 20D High Breakout: 68% win rate AND: High frequency (1,420 signals/year) Fast resolution (6.2 days average) Works across all regimes Simple and reliable to code Scanner value = frequency × regime-independence × win rate W-Pattern is excellent for manual trading but lower ranked for automated scanner

Detailed Report Per Setup (top 10 get full report)

SETUP #1 — 20D HIGH BREAKOUT + VOLUME 1.5x Exact conditions: Close > highest high of last 20 bars + Vol >= 1.5x avg + Close > 20 EMA Total signals: 2,847 over 2 years | Win rate 1.5R: 68% overall Bull: 76% | Neutral: 61% | Bear: 44% Cap: Large 72% Mid 68% Small 61% Best conditions: Bull + sector above 20D EMA → 79% win rate Volume sensitivity: 1.3x=61% | 1.5x=68% | 2x=74% → minimum 1.5x recommended ATR stop: captured 91% of wins | 20 EMA stop: captured 88%

SLOT ARCHITECTURE — 4 PRESERVED SLOTS

Slot	Type	Purpose	Filled By
Slot 1	LIVE	Signals sent to Telegram	Top ranked setup from backtest
Slot 2	LIVE	Signals sent to Telegram	2nd ranked setup from backtest
Slot 3	RESEARCH	Tracked silently — not sent	3rd ranked setup — monitoring
Slot 4	RESEARCH	Tracked silently — not sent	4th ranked setup — monitoring

■ You select which setups go in which slots AFTER seeing backtest results. Architecture is preserved — slots accept any setup that follows the standard interface.

Plug-In Interface — Any Setup Works

```
class Setup:
    name: str # e.g. "20D High Breakout + Vol 1.5x"
    conditions: dict # exact detection rules
    entry_rule: str # e.g. "close of breakout candle"
    sl_rule: str # e.g. "ATR 1.5x below entry"
    t1_rule: str # e.g. "1.5R"
    t2_rule: str # e.g. "2.5R"
    timeframe: str # "daily" or "weekly"
    regime_filter: str # "all" or "bull_only"
    def detect(df_daily, df_weekly) -> bool:
        # Returns True/False — same interface for every setup
        # Slot just calls detect() on whatever is plugged in
        # Swap setup = change JSON file + no code change needed
```

Slot Management Rules

PROMOTE: Research setup promoted to live when: Live tracking matches backtest win rate (+/- 10%) for 60+ signals Outperforms current live slot 1 or 2
DEMOTE: Live setup demoted to research when: Live win rate < backtest - 15% for 3 consecutive months
BETTER-RANKED: Better-ranked setup available from latest re-rank
REPLACE: Quarterly re-rank may surface new top setup
NEW: New setup tested in research zone first (slot 3 or 4) Promoted to live only after live tracking confirms backtest

LIVE SCANNER — SIGNALS FROM TOP 2 SETUPS

Same infrastructure as System 1. Hard filters identical. Global bleeding identical. Telegram delivery identical. Only difference: setups are data-selected not assumption-based.

Component	System 1	System 2
Stock universe	Same — ~450 NSE stocks	Same
Hard filters (5 layers)	Same — weekly gate, bleeding, corp actions	Same
Global bleeding filter	Same — GIFT + 70% basket	Same
Setups generating signals	Assumption-based (EMA cross, W-pattern, BB)	Data-selected (top 2 from backtest)
Conviction levels	4 levels (High/Mod/Low/Watch)	Same framework, backed by actual win rates
Hourly execution plan	YES — for top conviction signals	YES — same
News scanner (final 15)	YES — advisory for your decision	YES — same
Sector distribution cap	Max 2 per sector	Same
Telegram delivery	5 messages per scan	Same format
Research journal	Failure analysis + calibration	Same + backtest performance tracking

Signal Card — System 2 Difference

System 1 card says: ■■■■ HIGH CONVICTION (subjective estimate) System 2 card says: ■ SETUP: 20D High + Vol 1.5x
Backtest win rate: 68% (2,847 signals, 2 years) Current regime: BULL (historical in BULL: 76%) Live tracking: 71% (last 60 signals) ← updates over time This is the key difference: probability grounded in actual data

REPORTING STRUCTURE — SYSTEM 2

Weekly Report (Sunday 10 AM — 6 Messages)

Msg	Content
1	Live setup signals this week: count + outcome tracking (prior signals resolved)
2	Research zone signals: would setup 3+4 have worked? (tracked silently, not traded)
3	Live vs backtest comparison: are real outcomes matching predicted win rates?
4	Failure analysis: categorised failures + common factors + suggested tweaks
5	Research zone readiness: are setups 3+4 ready to promote based on live tracking?
6	Next week outlook: stocks near setup conditions + market regime assessment

Monthly Report (1st Sunday of each month)

Live setup performance vs backtest prediction: Are outcomes diverging from backtest? Why? (regime change? data drift?) Volatility in win rate: consistent or erratic? Research zone analysis: 60+ live signals tracked for each research setup? Win rate within +/- 10% of backtest? → candidate for promotion Significantly below backtest? → consider replacing with next ranked setup Backtest incremental update: rolling 2 years updated with latest month data Any rank changes? Any new setups entering top 10?

Quarterly Report (Every 13th Sunday)

SECTION 1: Full 2-year rolling re-rank – all 47+ setups New data replaces oldest quarter – rankings may shift SECTION 2: Top 2 live setup tweak suggestions Based on live performance data (3 months minimum) Specific condition adjustments with evidence: e.g. "Changing volume threshold from 1.5x to 1.8x would improve win rate from 64% to 71% based on last 90 signals in current market phase" SECTION 3: Research zone promotion decisions Setup 3: promote / continue / replace? Setup 4: promote / continue / replace? SECTION 4: System 2 vs System 1 comparison Which system found better setups? Which conviction levels / backtest win rates are more accurate? Recommendation: merge best of both into next version?

SYSTEM 2 — GITHUB ARCHITECTURE

```
setup-discovery-engine/ (PUBLIC repo – separate from System 1) ■■■■ .github/workflows/engine.yml ■■■■ data/ ■■■■
universe/ (same CSVs as System 1 – copy from there) ■■■■ sectors/ sector_map.csv ■■■■ corporate_actions/
actions.csv ■■■■ holidays/ nse_holidays files ■■■■ active_signals/ active_signals.csv ■■■■ backtest/ ■■■■
ranking.csv ← master setup ranking table ■■■■ setup_details/ ← one detailed CSV per setup ■■■■
regime_breakdown.csv ← performance by regime per setup ■■■■ live_setups/ ■■■■ slot1_setup.json ← selected live
setup 1 (you fill after backtest) ■■■■ slot2_setup.json ← selected live setup 2 ■■■■ slot3_research.json ←
research setup 3 ■■■■ slot4_research.json ← research setup 4 ■■■■ research_journal/ ← same as System 1 ■■■■ src/
■■■■ universe.py data_fetcher.py filters.py (same as S1) ■■■■ sector_bleeding.py global_markets.py market_regime.py
(same) ■■■■ indicators.py news_scanner.py parallel_runner.py (same) ■■■■ backtest_engine.py ← CORE: tests all 47+
setups ■■■■ setup_ranker.py ← ranks by win rate × regime × frequency ■■■■ setup_registry.py ← manages 4 slots
(live + research) ■■■■ live_scanner.py ← runs slot 1+2 live signals ■■■■ research_tracker.py ← tracks slot 3+4
silently ■■■■ research_journal.py ← same as System 1 ■■■■ weekly_reporter.py ← 6-message weekly report ■■■■
monthly_reporter.py ← live vs backtest comparison ■■■■ quarterly_reporter.py ← re-rank + tweak suggestions ■■■■
telegram_bot.py ← same delivery as System 1 ■■■■ main.py ← live scan (slots 1+2) – same schedule as S1 ■■■■
backtest_run.py ← initial + quarterly backtest (run manually) ■■■■ weekly_report.py ← Sunday report ■■■■
requirements.txt
```

System 2 Schedule

Run	Frequency	How	Time
Initial backtest	Once — first time	Manual: python backtest_run.py	20-40 min (one-time)
Live scan	Mon-Fri 5x + Sunday	GitHub Actions (same as S1)	~3.5 min/run
Monthly backtest	Monthly	GitHub Actions 1st Sunday extra	~10 min
Quarterly re-rank	Every 13th Sunday	GitHub Actions (extended Sunday run)	~30 min

First Run Workflow

STEP 1: Deploy setup-discovery-engine repo to GitHub STEP 2: Run backtest manually: python backtest_run.py (20-40 minutes – coffee break) STEP 3: Backtest sends ranking table to your Telegram channel STEP 4: You review ranking + detailed reports STEP 5: Select top 2 for live | Next 2 for research STEP 6: Fill slot JSON files in data/live_setups/ STEP 7: Enable GitHub Actions – live scanning starts STEP 8: Signals flow to Telegram channel automatically

System 2 Rating

Component	Rating	Notes
Backtest engine	8.5/10	47+ setups × 500 stocks × 2 years = comprehensive
Ranking methodology	8.0/10	Win rate × regime independence × frequency
Slot architecture	8.5/10	Plug-in design — easy to swap setups
Research zone tracking	8.0/10	Evidence-based promotion/demotion
Monthly live vs backtest	8.5/10	Catches data drift early
Quarterly re-rank + tweaks	9.0/10	Continuously improving system
OVERALL	8.5/10	Professional level — evidence-driven

COMBINED — HOW THE TWO SYSTEMS WORK TOGETHER

Relationship Between Systems

	System 1	System 2
Philosophy	Assumption-based setups	Data-driven setup selection
Purpose	Live signals now — you learn	Evidence first — validate assumptions
Setups	EMA cross, W-Pattern, BB (assumed)	Top ranked from backtest (proven)
Start	Immediate — deploy and run	After initial backtest run (1 session)
Signals	To Telegram now	To Telegram after setup selection
Value now	Practice + learning + research journal	Backtest reveals which setups actually work
Value later	Upgrade with System 2 findings	Eventually replaces System 1 assumptions
Research	Failure analysis + calibration	Live vs backtest + re-rank + tweaks
Shared	Universe, filters, bleeding, news, delivery, journal	—

Recommended Path

MONTH 1-3: Run BOTH systems simultaneously System 1: generates signals based on assumptions System 2: runs initial backtest → reveals ranked setups Compare: do System 1 setups appear in System 2 top rankings? MONTH 3-6: System 2 has 60+ live signals per setup tracked Live win rates emerging – matching backtest? System 1 research journal: which conviction levels actually predict outcomes? MONTH 6+: Quarterly re-rank done – rankings updated with live data Decision: keep System 1 as-is OR upgrade setups with System 2 findings Potentially: best setups from System 2 → added to System 1 architecture System becomes genuinely data-driven over time THIS IS THE PROFESSIONAL IMPROVEMENT LOOP: Assumption → Test → Evidence → Upgrade → Repeat

DISCLAIMER: Both systems are for educational and personal research only. Not SEBI registered investment advice. All signals are for study purposes only. All trading decisions are the sole responsibility of the user. Past performance does not guarantee future results. Always use stop losses.